



CERTIFICATE OF APPROVAL

No CF 761

This is to certify that, in accordance with
TS00 General Requirements for Certification of Fire Protection Products
The undermentioned products of

CENTRAL VENTILATION SYSTEMS

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Have been assessed against the requirements of the Technical Schedule(s)
denoted below and are approved for use subject to the conditions
appended hereto:

CERTIFIED PRODUCT

SAFE4 Non-Coated Fire
Resistant Steel Ductwork System

TECHNICAL SCHEDULE

TS48 Smoke Control and Fire
Resisting Ductwork Systems

Signed and sealed for and on behalf of Warringtonfire Testing and Certification Limited

Paul Duggan
Certification Manager



Issued:
Reissue:
Valid to:

29th June 2010
05th August 2024
6th May 2028





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Manufacture:

The following addresses manufacture the products listed on this certificate and the FPC has been conducted to ensure the product meets the scheme requirements:

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SAFE4 Non-Coated Fire Resistant Steel Ductwork Systems

1. This approval relates to the use of the above non-coated ductwork systems in providing fire resistance of up to 240 minutes stability and integrity (duct A fire attack from outside the duct and duct B fire attack from inside the duct) and up to 180 minutes insulation (duct A) and up to 60 minutes insulation (duct B), as defined in BS 476: Part 24: 1987 (ISO 6944: 1985), along with providing 4 hour smoke extract, duct A, and 2 hours smoke extract duct B performance.
2. This certification is provided to the client for their own purposes and we cannot opine on whether it will be accepted by Building Control authorities or any other third parties for any purpose.
3. The ductwork systems are approved on the basis of:
 - I. Initial type testing
 - II. Audit testing at the frequency specified in TS48
 - III. A design appraisal against TS48
 - IV. Certification of quality management system.
 - V. Inspection and surveillance of factory production control
4. The ductwork systems comprise non-coated SAFE4 steel ductwork. If insulation performance is required the SAFE4 ductwork is fitted with Insulfrax ® S / LT/ LTX Blanket around the outside of the ductwork with thickness as indicated in this certificate.
5. This approval is applicable to insulated and uninsulated SAFE4 steel ductwork systems as described within this Certificate.
6. The ductwork systems shall be mechanically supported from floor and/or wall constructions or structural steel members having a fire resistance of at least the same period as the ductwork systems.
7. The approval relates to on going production. Product and/or its immediate packaging is identified with the manufacturers' name, the product name or number, the CERTIFIRE name or name and mark, together with the CERTIFIRE certificate number and application where appropriate.
8. SAFE4 ductwork is a non-coated system that does not require any coating treatment to achieve the fire resistance rating as certified within this certificate



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This certification is based on the following fire resistance test reports:

- 332925A
- 332925C
- 17104C
- 17148C
- 17245C
- 17321C
- 17450B
- 17450D
- 17451B

A handwritten signature in black ink, appearing to read "Paul Rogers".



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Scope: Mild steel, galvanised steel or stainless steel ventilation, smoke extract, kitchen and pressurisation ducts with accessories e.g. Sound Attenuators and Access Doors etc. providing fire resistance of up to 240 minutes stability and integrity (duct A fire attack from outside the duct and duct B fire attack from inside the duct) and up to 180 minutes insulation (duct A) and up to 60 minutes insulation (duct B), as defined in BS 476: Part 24: 1987 (ISO 6944: 1985, with the limitations detailed within this report, along with providing 4 hour smoke extract, duct A, and 2 hours smoke extract duct B performance.

Orientation: Horizontal and vertical.

Shape: Rectangular circular, and flat oval

Size: Up to 6000mm wide x 3000mm high

Construction details are provided in tables within Appendix A along with installation details in Appendix B.

Rectangular Steel Duct Assembly (RGIL120 & RGIL240, RSSL120 & RSSL240)

The steel duct is constructed in sections up to 1500mm long using hot dipped galvanized steel sheet, mild steel sheet or stainless-steel sheet, either 0.8 mm, 1.0 mm, 1.2 mm or 1.4 mm thick. The sections are constructed with Pittsburgh lock, grooved corner (RAS), fully welded longitudinal seams and flanged cross-joints. Where construction requires a heavy gauge system incorporating 1.50 mm or 2.0 mm thick steel, then the duct specifications are incorporated in HGDS120 and HGDS240.

At each cross joint fire rated SAFE4 slip-on flanges are fastened to the duct with either 4.8mm-diameter steel rivets at 150mm nominal centres or spot welds at 100mm nominal centres. SAFE4 slip-on flanges are fitted with steel corner plates of the respective flange size that slot into the flanges at each corner and are bolted together with an M10 steel nut and bolt at each corner. The cross joint flanges are fastened together with M8 bolted steel clamps at 250mm maximum centres. Where RSA flanges are used the flanges are bolted together at each corner and at 250mm maximum centres with minimum M8 steel nuts and bolts. Equivalent J2, J3, J4, J5 & J6 rated slip-on flanges or RSA flanges are used for the respective rating. Each cross-joint is sealed with Arboseal Intumescent Tape or Pyrosafe Tape GT500, 15mm wide x 3mm thick fitted between the mating flanges. All other duct joints are sealed with Arbo XL1075, Pyrosafe FS1075 or Nullifire M701 fire rated intumescent sealant.

For ducts with stiffening collars the stiffening collar is fastened to all four sides of the duct using 4.8mm-diameter steel rivets at 150mm nominal centres or spot welded at 100mm intervals. For



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ducts with a longer side greater than 1000mm, the stiffeners, steel angle minimum 50mm x 40mm x 1.5mm thick and fastened to the duct through the 40mm leg, are bolted, riveted or welded together at the corners. Alternative S3 rated stiffening collars are fitted in the same way. The size of the angle is increased to minimum 50mm x 50mm x 5mm thick for 120 minute ducts over 2500mm longer side and 240 minute ducts over 2000mm longer side. Alternative S5 rated stiffening collars may be fitted in the same way.

For duct sizes over 1000mm a steel tie rod type stiffener is fitted at one end of each duct section, at mid width, to the flanges of the cross joint. The minimum M10 threaded rod is bolted or welded to the cross joint. Two tie rods, equally spaced across the width of the duct, are fitted for duct widths greater than 2000mm.

All duct piercings, fastenings or locks are sealed with Arbo XL1075, Pyrosafe FS1075 or Nullifire M701 fire rated intumescent sealant..The above construction details relate to commercial grade steel.

Details of the duct constructions for sizes up to 3000mm wide x 2000mm high are summarised in Appendix A construction tables. These give details of the requirements for steel thickness, cross joints and intermediate stiffeners, along with hanger/bearer details.

Mild steel welded ducts (RMSW120 & RMSW240)

The steel duct is constructed in sections up to 1500mm long using cold rolled or hot rolled sheets a minimum of 1.4mm thick. Longitudinal joints are fully welded end-to-end. Where construction requires a heavy gauge system incorporating 1.50 mm or 2.0 mm thick steel, then the duct specifications are incorporated in HGDS120 and HGDS240.

At each cross joint rolled steel angles (RSA) are fully welded to each end of the duct. The minimum sizes of RSA for various duct sizes up to 3000mm are shown in Tables A3 & A4. The RSA flanges are bolted together at each corner and at 250mm maximum centres with minimum M8 nuts and bolts. Each cross-joint is sealed with Arboseal Intumescent Tape or Pyrosafe Tape GT500, 15mm wide x 3mm thick fitted between the mating flanges. Alternatively, cross-joints may be welded together to form a fully welded construction. All other duct joints are sealed with Arbo XL1075, Pyrosafe FS1075 or Nullifire M701 fire rated intumescent sealant.

The stiffening collar is fitted to all four sides of the duct with MIG stitch welds at 200mm nominal centres and welded together at the corners, with the same size and tie bar requirements as galvanised steel ducts. The minimum sizes of stiffening collar for various duct sizes up to 3000mm are shown in Appendix A construction tables.



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Construction details used for welding Stainless steel ducting are shown in construction tables
RSSW120 & RSSW240

All duct piercings, fastenings or locks are sealed with Arbo XL1075, Pyrosafe FS1075 or Nullifire M701 fire rated intumescent sealant.

Details of the duct constructions for sizes up to 3000mm wide x 2000mm high are summarised in Appendix A construction tables. These give details of the requirements for steel thickness, cross joints and intermediate stiffeners, along with hanger/bearer details.

Multi-panel ducts (MSTM)

The steel duct is constructed in sections up to 1500mm long using hot dipped galvanised steel sheet, mild steel sheet or stainless steel sheet 1.2mm thick. The duct comprises corner panels, maximum 1000mm x 1000mm, and flat panels, maximum 2000mm wide. 'U-panels' may be constructed a maximum width of 1500mm and a maximum height of 1000mm. Each panel is reinforced around the perimeter and at mid-depth with steel angle, 50mm x 50mm x 5.0mm thick. The angles are fastened to the sheet with M6 steel bolts or 4.8mm steel rivets at maximum 150mm centres or spot welds at 75mm maximum centres. The panels are fastened together with M10 steel setscrews and nuts at 250mm maximum centres. Each section is fitted with tie bars/drop rods at 1000mm maximum centres with the two end rods/bars positioned nominally 250mm from ends of the section. The bars are connected to the angle at mid-depth with a nut and washer on both sides. The bars/rods are extended to the supporting structure above the duct to form the hangers. The spacing of the hangers and the size of the steel hanger components must be adjusted so that the tensile stress in the bars/rods does not exceed 15N/mm² for fire resistance periods up to 60 minutes, 10N/mm² for fire resistance periods up to 120 minutes and 6N/mm² for fire resistance periods up to 240 minutes.

The maximum spacing of the hangers is 1500mm. The duct sections are fastened together with M10 steel set screws and nuts at 250mm centres. Each cross-joint is sealed with Arboseal Intumescent Tape or Pyrosafe Tape GT500, 15mm wide x 3mm thick fitted between the mating flanges. All other duct joints are sealed with Arbo XL1075, Pyrosafe FS1075 or Nullifire M701 fire rated intumescent sealant. The maximum size of the steel ducts is 6000mm wide x 3000mm high.



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Insulation

The insulation material fitted around the ducts is Insulfrax ® S / LT / LTX blanket. The AES mineral wool blanket has a nominal density of 96kg/m³. The blanket, supplied in rolls 610mm or 1220mm wide, is provided in thicknesses of 38 and 50mm. The thicknesses of insulation required for the ducts are shown in Table 1.

Table 1 Insulation thickness

Fire resistance - minutes	Protection thickness – mm	
	Duct A	Duct B
60	38	76
120	76	-
180	100	-

The steel duct is clad with one or two layers of Insulfrax ® S / LT / LTX blanket that are wrapped around the duct. For two layer systems all joints are tight butt joints. Joints are staggered between layers by 300mm. For single layer protection all joints are overlapped by nominally 75mm. The Insulfrax ® S / LT / LTX blanket is fastened to the duct with 2mm-diameter steel pins and spring steel washers, minimum 38mm diameter. The pins are welded, screwed or riveted to the steel duct at 200mm centres across the width of the duct and 200mm centres along the length of the duct. On the bottom face of horizontal ducts the ends of the pins are bent over by 90° once the protection and washers have been fitted. The arrangement is shown in Figures 3 & 4.

Hanger supports

Details of the hanger supports for different sizes for fire exposure up to 240 minutes are given in the Tables A17, A18, A19, A20, A21 & A22 for each type of duct construction. Each steel hanger consists of two threaded drop rods, minimum M8, and a channel section bearer. The bearer is fastened to the drop rods with hexagonal nuts and washers. The spacing of the hangers and the size of the steel hanger components must be adjusted so that the tensile stress in the rods and the bending stress in the angles/channels do not exceed 15N/mm² for fire resistance periods up to 60 minutes, 10N/mm² for fire resistance periods up to 120 minutes and 6N/mm² for fire resistance periods up to 240 minutes. The maximum spacing of the hangers is 1500mm. Each hanger must be positioned so that it is adjacent to (within 100mm of) a cross joint or a stiffening collar. The Unistrut bearers may be fitted within or outside the insulation. Larger bearers are



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fitted outside the insulation. The drop rods are positioned outside the insulation material but at a distance of not more than 50mm from the insulation.

The fixings used to fasten the threaded rod hangers to the structure must be suitable for the type of substrate and installed in accordance with the manufacturer's instructions for the required fire performance and loading.

When fixing to a steel supporting construction, a hole should be drilled through the steel member, allowing the drop rod to be supported by a steel nut and washer above. If a clamp type fixing is used the clamp must be composed of steel, suitable for the purpose and pass around the steel member and be fastened back on itself. Clamps that rely on friction to hold them in place are not suitable.

A supporting concrete construction must consist of steel reinforced structural concrete and be capable of supporting the anticipated loads for the required period of fire resistance. It is not possible to specify a minimum grade of concrete due to the wide range of different types of concrete (e.g. dense aggregate, lightweight aggregate, autoclaved aerated). However, in all cases the type of fixing should be designed for the type of concrete and the depth of the fixing may have to be increased to maintain adequate engagement.

Penetration seal

The penetration seal system, where the duct passes through fire compartment walls or floors, is the same as that tested in report WF No. 332925A for uninsulated ducts and the same as that tested in report WF No. 332925C for insulated ducts. Safire coated mineral fibre slab, minimum 50mm thick x 160kg/m³ density for uninsulated duct, or Firetherm coated Fire Stop Board, minimum 100mm thick x 140kg/m³ density for insulated ducts, is fitted around the duct within the opening. All edges between the slab, the wall or floor and the steel duct are sealed with Nullifire M701 mastic on both faces of the penetration seal.

A steel plate collar, minimum 1.0mm thick, is fitted around the steel duct on both sides of the wall, overlapping the wall or floor by at least 100mm on each side. The plate is fixed to the wall or floor with M8 all-steel expanding anchors at 300mm maximum centres. All edges between the plate collar, the wall or floor and the steel duct are sealed with Alfacryl intumescent acrylic sealant. A duct cross joint or a steel stiffening collar must be positioned within the thickness of the seal assembly.

Alternative penetration seal other than detailed or tested should not be used.



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One, two and three sided ducts (OSTM, DSTM & TSTM)

One, two and three side ducts are formed by attaching the steel duct directly to the adjacent concrete floor and/or concrete/masonry wall. The longitudinal edges of each duct section are formed with a flange as specified in relevant construction schedule for four-sided ducts. The flanges are fastened to the floor and/or wall with M10 all-steel expanding anchors, with steel washers, at 350mm maximum centres. Arboseal Intumescent Tape or Pyrosafe Tape GT500, 15mm wide x 3mm thick, is fitted between the flanges and the floor or wall to seal the joints. No hangers are required. All other aspects of the duct construction are the same as for four sided ducts. The construction details for one, two and three sided ducts are shown in Figures 5 to 7.

Vertical ducts

Vertical duct installation is shown in Figure 2.

The construction of vertical ducts is the same as for horizontal ducts. In order to support the duct system in the vertical orientation, steel sections are fastened to the steel duct, either along the two longer sides or on all four sides. The steel sections either span across the opening in the concrete floor or form part of a cantilever bracket that is fastened to the building structure. The maximum vertical distance between supports is 4.0m. Also, to prevent buckling of the duct, the distance between supports should not exceed 8 times the smallest lateral dimension across the outside face of the steel duct. The weight of the duct system is taken by the building structure at each floor level. The stress limit within the steel supports is 10N/mm² for fire exposures of up to 120 minutes and 6N/mm² for fire resistance periods up to 240 minutes. On insulated ducts, the supporting steel members, where they emerge from the protection around the duct, are clad with the same thickness of insulation for at least 300mm.

The design of the penetration seal for vertical ducts through concrete floors and the application of the Insulfrax ® S / LT/ LTX blanket are the same as for horizontal ducts.



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Smoke outlet ducts

If the duct is to act as a Type B (fire inside) smoke outlet duct, then the smoke extract ductwork shall be further reinforced by 2 internal 10mm stiffening rods, fitted with steel nuts and washers, spaced equally across the stiffening collar, along with an additional tie bar at the cross joint to fulfil the additional criteria of maintaining 75% of cross section area of the duct as required by BS476:Part 24 to maintain adequate extraction. This construction has been shown to provide 2 hours smoke extract performance.

Where type A (Fire outside) smoke extract performances required, such as pressurisation systems, report 332925C showed 4 hour smoke extract performance with the standard duct ventilation design.

Kitchen extract ducts

The construction of kitchen extract ducts is the same as for normal ventilation ducts for stability and integrity, the insulated kitchen extract duct systems require an insulation thickness of 100mm Insulfrax S blanket /LT/ LTX blanket for a 30-minute rating. Where a longer insulation rating is required, Insulfrax S / LT/ LTX blanket does not suffice.

Circular and flat oval ducts (CGIL120, CGIL240, CSSL120, CSSL240, CSSW120, CSSW240, CMSW120 & CMSW240)

Details of the construction of circular and flat oval straight seamed and spirally wound steel ducts are generally in accordance with those described in DW/144 - Specification for sheet metal duct work - low, medium and high pressure/velocity air systems (published by the Building & Engineering Services Association), or equivalent specification, but with the following minimum requirements:

Circular ducts

- The thickness of steel sheet used in the construction of the ducts is 0.8mm for ducts up to 1000mm diameter, 1.0mm for sizes up to 1250mm..
- The longitudinal seams and cross joints must be at least equivalent in strength to the seams used on the tested rectangular duct.
- The spiral seam of spirally wound ducts must be equivalent to the grooved seam.
- The maximum diameter of circular ducts covered by this assessment is 1250mm.



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- The design of the penetration seal where the duct passes through fire compartment walls or floors must be equivalent to the designs specified for rectangular ducts.

The design of the hanger support systems should be such that the maximum stress in the hanger components does not exceed 15N/mm² for fire resistance periods up to 60 minutes, 10N/mm² for fire resistance periods up to 120 minutes and 6N/mm² for fire resistance periods up to 240 minutes.

Details of the duct construction at various sizes are given in Tables A9 to A16. These give details of the requirements for steel thickness, cross joints and intermediate stiffeners. All duct piercings, fastenings or locks are sealed with Nullifire M701 fire rated intumescent sealant.

The Insulfrax ® S / LX/ LTX blanket insulation is fitted to circular ducts in the same manner as for 4-sided rectangular ducts using steel pins. The pins are fitted in a grid of 200mm along and around the duct.

Flat oval ducts

Minimum steel thickness for flat oval ducts:

Type of duct	Maximum surface area per metre length – m ² /m			
	≤1.2	≤2.0	≤2.6	≤3.83
Ventilation	0.8	1.0	1.2	1.2
Smoke outlet/extract	1.0	1.0	1.2	1.2
Kitchen extract	1.0	1.0	1.2	1.2

For flat oval ducts over 500mm wide x 150mm deep, tie rods, with a minimum diameter of 12mm, must be fitted between the top and bottom walls of the flat portion of the duct, as follows:

- for ducts between 500mm wide and 700mm wide, there should be one tie rod located at mid-width of the duct at 1000mm centres along the duct.
- for ducts between 701mm wide and 900mm wide, there should be one tie rod located at mid-width of the duct at 750mm centres along the duct.
- for ducts between 901mm wide and 1200mm wide, there should be two tie rods equi-spaced across the duct at 500mm centres along the duct.
- for ducts between 1201mm wide and 1800mm wide, there should be tie rods at 250mm centres along the duct, alternating between one tie rod centrally located and two tie rods equi-spaced across the duct.



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All duct piercings, fastenings or locks are sealed with Arbo XL1075, Pyrosafe FS1075 or Nullifire M701 fire rated intumescent sealant.

All hanger/bearer details are the same as for the rectangular ducts.

Access panels (*RAP1, RAPB & RAPW*)

The details of the construction of three different designs of access panel are shown in Figures 10 to 12. The maximum opening size of the panels is 600mm x 400mm. The panels must not interrupt a cross joint or a stiffener.

Access door testing is not covered under BS476 Part 24; however, the proposed system uses the same insulation as the ducting to minimise any insulation loss caused by the use of the access panel. Alternatively, access doors can be installed as tested within the EN1366-1 test reports, such as 17148C, 17245C, 17321C, 17450D, 17451D.

Although the access panels have not been tested, the designs of the panels are robust. The opening in the duct wall is reinforced around the perimeter with steel framing members and the panel is held in position with steel bolts or fasteners at close centres. A ceramic tape gasket is used to seal the panel to the duct. This type of seal has been shown to be suitable for fire exposures up to 240 minutes.

The access panel is fitted into a 40mm flanges x 1.5mm thick channel section frame that is fastened to the duct skin at 200mm maximum centres with 4.8mm steel rivets or M6 bolts and nuts and sealed with a bead of Arbo XL1075, Pyrosafe FS1075 or Nullifire M701 fire rated intumescent sealant. Alternatively, the channel section frame may be welded on to the duct. The web depth of the channel matches the insulation thickness. A 25mm-wide ceramic tape is fastened to the panel side of the frame with 4.8mm steel rivets and sealed with a bead of intumescent sealant. The access panel is made of two 1.2mm-thick steel trays filled with rock wool of 48kg/m³ nominal density. The two trays are fastened together with spot welds at 100mm nominal centres. The thickness of the panel can vary from 25mm to 75mm.

The panel is held in the frame with M6 steel wingnuts and washers fitted onto the M6 screws welded in frame. The fasteners are fitted on all four sides of the panel at 200mm maximum centres.

The single skin bolted panel comprises 1.2mm steel sheet with a 5mm lip formed on all four edges of the panel. A door handle, made of 10mm steel bar, is welded to the panel. The access panel is fitted into a 40mm x 1.5mm thick channel section frame that is fastened to the duct skin



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at 200mm maximum centres with 4.8mm steel rivets or M6 bolts and nuts and sealed with a bead of Arbo XL1075, Pyrosafe FS1075 or Nullifire M701 fire rated intumescent sealant. M6 steel screws are welded to the channel section frame at 200mm maximum centres. The panel is fastened to the channel frame with steel wingnuts and washers fitted onto the M6 screws. A 25mm-wide ceramic tape is fastened to panel side of the frame with 4.8mm steel rivets and sealed with a bead of intumescent sealant. Where the panel is required to sit flush with the duct, the channel section may be replaced with a 50mm x 5.0mm thick flat steel bar that is bolted or riveted to duct wall at 200mm maximum centres.

The single skin welded panel comprises 1.2mm steel sheet with a 5mm lip formed on all four edges of the panel. A door handle, made of 10mm steel bar, is welded to the panel. The frame of the panel assembly consists of steel flat bar, 50mm wide x 5mm thick, with M6 steel screws welded to the flat bar frame at 200mm maximum centres. The flat bar frame is welded to the duct skin. The panel is fastened to the frame with steel wingnuts and washers fitted onto the M6 screws. A 25mm-wide ceramic tape is fastened to door side of the frame with 4.8mm steel rivets and sealed with a bead of intumescent sealant.

Sound attenuators (RSAA)

The construction of the body of the attenuator units is at least as strong as the SAFE 4 fire resistant duct system. The presence of the pods within the body of the attenuators adds to the structural strength of the assemblies and helps to maintain the cross-sectional shape of the duct. The additional weight of the pods must be allowed for when calculating the maximum stresses within the hanger supports. The general design of the attenuator performance has been shown during testing where pods were included in the final section of duct construction. Steel pods are fitted within the attenuator casing. Each pod is constructed with bull nose sections at each end, minimum 0.8mm thick, and perforated sides, minimum 0.7mm thick. The pods are filled with mineral wool with a minimum density of 48kg/m³. The pods are fastened to the body of the attenuator with 4.8mm steel rivets at 150mm nominal centres or spot welds at 100mm centres.

The sound attenuator units may be insulated with the same insulation system as used for the ventilation duct system. Details of construction are shown in Figure 12. The attenuators may be constructed with removable baffles held in place with RSA welded, bolted or riveted to the casing at 200mm centres.

The sound attenuator units must each be supported by at least two steel hangers in the same manner as the steel duct system. The spacing of the hangers and the size of the steel hanger components must be adjusted so that the tensile stress in the rods and the bending stress in the channel bearers do not exceed 15N/mm² for fire resistance periods up to 60 minutes, 10N/mm²



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for fire resistance periods up to 120 minutes and 6N/mm² for fire resistance periods up to 240 minutes.

Internally flanged and stiffened ducts (RSTD-I)

Where it is not possible for the duct flanges and stiffeners to be fitted to the outside the duct, RSA flanges to meet cross joint ratings and stiffeners to meet stiffener ratings can be fitted internally. The RSA profiles are fastened to the duct in the same way as for external profiles. The profiles are welded, bolted or riveted at corners in the same way as they are fastened when fitted externally. Construction details are shown in Figure 8.